

8th Grade Unit 14 Assessment Guide

General Information	Unit 14 builds on student understanding of probability from grade 7. As this unit contains unique content, it was strategically placed at the end of the year so that student learning of these benchmarks is close to the statewide assessment. This eliminates the need to review content that does not have a strong connection with other concepts students have learned in grade 8. The assessment asks students to give their answers using a variety of forms. When reviewing student work, consider minor deductions if students have the correct answer but an incorrect form. Question 2a is aligned with two benchmarks. This was done because both benchmarks would require the student to determine the number of outcomes, yet neither benchmark states this specifically. It is a scaffolded step to both benchmarks and may help teachers determine if students are understanding how the total number of outcomes are determined for a repeated experiment. Question 6c is an example of a question that has a high DOK. Students will need to think about what would happen in an experiment given the two different sample spaces. Teachers may wish to monitor how their students complete each statement to consider any misconceptions students have. The assessment specifically presents students with new ways to view and relate to experiments. For example, a bar chart is provided in question 6. Reading and interpreting bar charts is prior knowledge, so students should have familiarity with using the presentation. If teachers find that many students struggle with new ways of seeing the content, they can encourage students by discussing the question after the assessment. Helping students take what they have learned and apply it to new contexts is important, but students may need coaching to help them gain confidence. Guiding students on how to read a question and to connect what is being asked for will help offset students being flustered by new contexts.
Calculator Information	The questions on the unit assessment are calculator neutral. Teachers can determine whether students should use a calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1a	Benchmark(s)	MA.8.DP.2.1
<i>P</i> represents the pitcher side of the coin <i>B</i> represents the batter side of the coin <i>{PPP, PPB, PBP, PBB, BPP, BPB, BBP, BBB}</i>		Recommended Scoring (16 Total Points)	Consider distributing the points as 2 points for each. Consider deducting a minor deduction of points if students use “heads” and “tails” for the outcomes instead of “pitcher” and “batter”.
Question	1b	Benchmark(s)	MA.8.DP.2.2
12.5%		Recommended Scoring (4 Total Points)	Consider no partial credit.
Question	2a	Benchmark(s)	MA.8.DP.2.1*, MA.8.DP.2.2*
289		Recommended Scoring (5 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	2b	Benchmark(s)	MA.8.DP.2.2
0.055		Recommended Scoring (5 Total Points)	Consider deducting a minor deduction of points if students give the probability as a percentage or a fraction.

Question	3	Benchmark(s)	MA.8.DP.2.2
$\frac{1}{36}$		Recommended Scoring (5 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	4	Benchmark(s)	MA.8.DP.2.2
$\frac{4}{81}$		Recommended Scoring (5 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	5a	Benchmark(s)	MA.8.DP.2.3
$P(\text{Sour Apple, Watermelon}) = 0.069$		Recommended Scoring (5 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	5b	Benchmark(s)	MA.8.DP.2.3
$P(\text{Watermelon, Cotton Candy}) = 0.083$		Recommended Scoring (5 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

Question	5c	Benchmark(s)	MA.8.DP.2.3
	☐ $P(\text{Cotton Candy, Cotton Candy})$ ☐ $P(\text{Sour Apple, Tutti – Frutti})$ ☐ $P(\text{Sour Apple, Watermelon})$ ☐ $P(\text{Tutti – Frutti, Cotton Candy})$ ☐ $P(\text{Watermelon, Cotton Candy})$	Recommended Scoring (15 Total Points)	Consider giving 5 points for each correct choice and deducting 5 points for one incorrect choice. Consider giving no points if students select all choices.
Question	6a	Benchmark(s)	MA.8.DP.2.3
	9.4%	Recommended Scoring (5 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	6b	Benchmark(s)	MA.8.DP.2.3
	$(A,Y); (G,R); (G,Y); (Y,A)$	Recommended Scoring (20 Total Points)	Consider giving 5 points for each correct outcome. Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

Question	6c	Benchmark(s)	MA.8.DP.2.3
If, in a second experiment, Elizabeth spun the spinner twice 32 times, she would expect the new results to	○ have experimental probabilities that are different than the first experiment. ○ have exactly the same experimental probabilities as in the first experiment. ○ have experimental probabilities that are closer to the theoretical probabilities. ○ have outcomes that are exactly $\frac{1}{2}$ of the outcomes in the first experiment.		
If, in a second experiment, Elizabeth spun the spinner twice 640 times, she would expect the new results to	○ have exactly the same experimental probabilities as in the first experiment. ○ have experimental probabilities that are closer to the theoretical probabilities than the first experiment. ○ have outcomes that are exactly 10 times the outcomes in her first experiment. ○ have experimental probabilities that are exactly twice as much as in the first experiment.	Recommended Scoring (10 Total Points)	Consider giving 5 points for each correct statement.